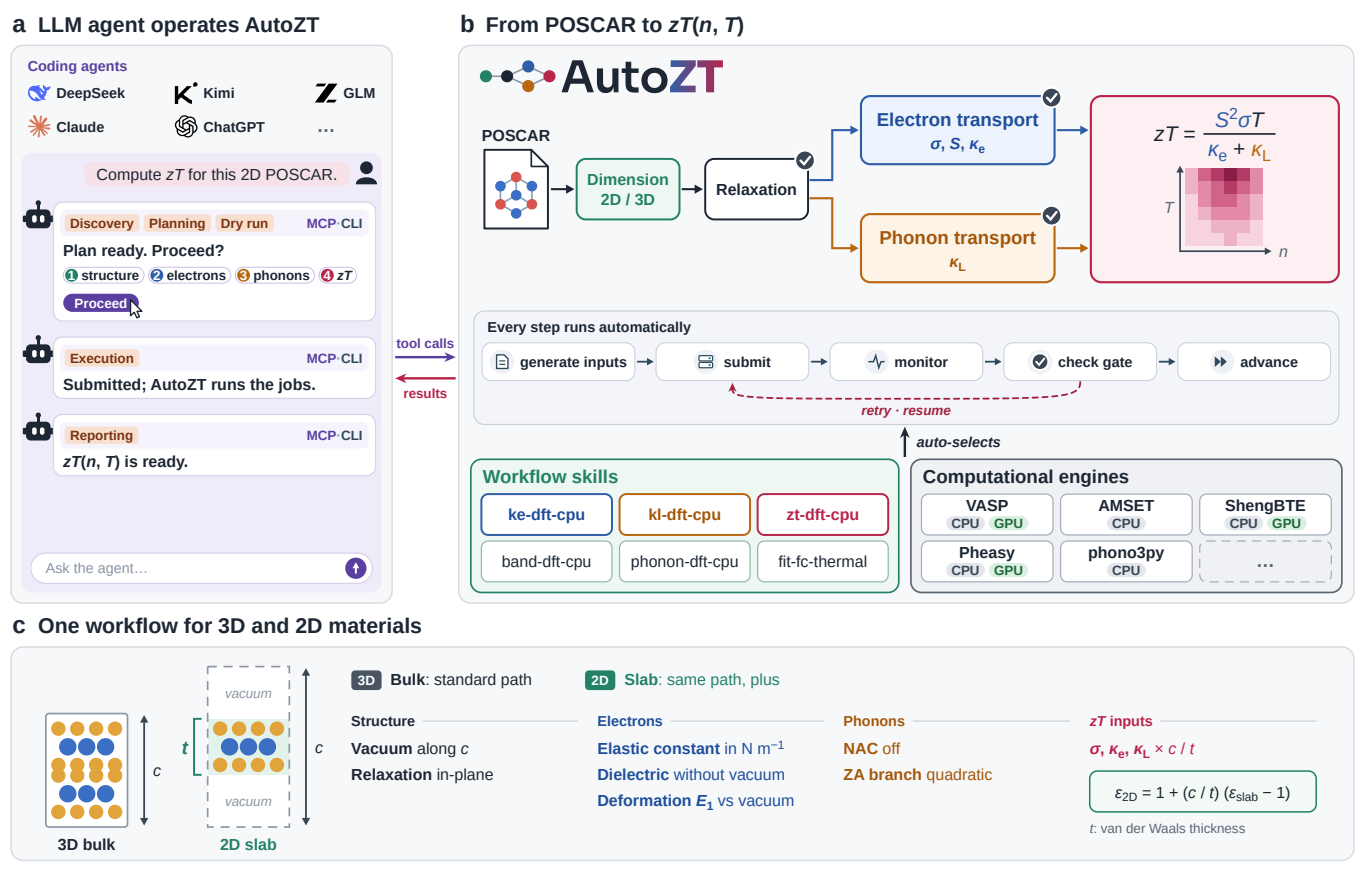

a LLM agent operates AutoZT
Coding agents
DeepSeek
Kimi
GLM
Claude
ChatGPT
...
Compute zT for this 2D POSCAR.
Discovery
Planning
Dry run
MCP·CLI
Plan ready. Proceed?
1 structure
2 electrons
3 phonons
4 zT
Proceed
Execution
MCP·CLI
Submitted; AutoZT runs the jobs.
Reporting
MCP·CLI
zT(n, T) is ready.
Ask the agent…
tool calls
results
b From POSCAR to zT(n, T)
AutoZT
POSCAR
Dimension
2D / 3D
Relaxation
Electron transport
σ, S, κe
Phonon transport
κL
zT = S²σT / (κe + κL)
T
n
Every step runs automatically
generate inputs
submit
monitor
check gate
advance
retry · resume
auto-selects
Workflow skills
ke-dft-cpu
kl-dft-cpu
zt-dft-cpu
band-dft-cpu
phonon-dft-cpu
fit-fc-thermal
Computational engines
VASP
CPU GPU
AMSET
CPU
ShengBTE
CPU GPU
Pheasy
CPU GPU
phono3py
CPU
...
c One workflow for 3D and 2D materials
c
t
vacuum
vacuum
c
3D bulk
2D slab
3D Bulk: standard path
2D Slab: same path, plus
Structure
Vacuum along c
Relaxation in-plane
Electrons
Elastic constant in N m⁻¹
Dielectric without vacuum
Deformation E1 vs vacuum
Phonons
NAC off
ZA branch quadratic
zT inputs
σ, κe, κL × c / t
ε2D = 1 + (c / t) (εslab − 1)
t: van der Waals thickness